Best tested Transformer-only challenger: -0.31 points

Primary question: can an all-Transformer ensemble match the current 15-model v4 incumbent?

Horizon	Model	Top-1	Top-3	F1 / 33	NLL	Brier	ECE
+1 s	v4 incumbent	67.09%	91.01%	0.4125	1.0688	0.4515	4.16 pp
+1 s	Transformer only	66.67%	91.23%	0.4119	1.0377	0.4484	2.63 pp
+1 s	Secondary blend	67.10%	91.47%	0.4171	1.0227	0.4439	3.51 pp
+2 s	v4 incumbent	66.62%	90.78%	0.4180	1.0855	0.4564	3.63 pp
+2 s	Transformer only	66.47%	90.98%	0.4130	1.0497	0.4515	2.34 pp
+2 s	Secondary blend	66.73%	91.20%	0.4169	1.0353	0.4474	2.94 pp
+5 s	v4 incumbent	64.86%	89.86%	0.3767	1.1243	0.4763	4.36 pp
+5 s	Transformer only	64.50%	90.14%	0.3639	1.0980	0.4744	3.82 pp
+5 s	Secondary blend	64.96%	90.39%	0.3680	1.0852	0.4699	4.71 pp

Transformer only: mean accuracy gain -0.31 pp; 95% scenario interval [-1.18, +0.64] pp.
Secondary blend: mean accuracy gain +0.07 pp; 95% scenario interval [-0.63, +0.84] pp.

Transformer only: NLL improved at all horizons; Brier improved at all horizons; ECE improved at all horizons.
Secondary blend: NLL improved at all horizons; Brier improved at all horizons; ECE changed across horizons.

Best tested is not an established optimum. Intervals are conditional on frozen models and previously inspected development data; an interval crossing zero does not prove equivalence. The secondary blend is a different model family.

What the Transformer variants actually change

Same observation information: 91 features and 20/60-second sparse histories, with all earlier origins retained

Family	20 s / 60 s	Representation	Classification
Pooled Transformer	26 / 66 tokens	Final state + masked mean + current state	Transformer
Temporal flattened	26 / 66 tokens	Keep all contextual token positions in readout	Transformer
Multiscale patch	9 / 17 patches	Five older tokens / two recent / current separate	Transformer
Attention + MLP	26 / 66 tokens	Attention plus a separate full-history MLP path	Hybrid

New attention variants encode known relative times and gaps, mask missing history, and use independently initialized pre-norm blocks. Bidirectional attention stays inside observations available at the forecast origin. The pooled comparison retains its original learned-position representation.

Primary: 60 s patch T + 60 s temporal T; 6 Transformer members, no incumbent members.

Secondary: 75% new (60 s patch T + 60 s temporal T) + 25% preserved v4; 21 total models.

Transformer-only means no separate MLP/TCN forecasting members: standard feed-forward blocks and current-state readouts remain part of the network. Width, regularization and representation are recipe choices, not a proof that attention alone caused any gain. Full settings and parameter counts accompany the PDF.

A bounded screen and confirmation inside original training

147 fit / 21 stop / 42 selection scenario groups per partition; all ten versions of each scenario stay together

Screen / reference	Family	Top-1	NLL	F1 / 33
v4 reference	15 models	60.53%	1.217	0.337
20 s pooled T	Transformer	56.11%	1.540	0.319
60 s pooled T	Transformer	55.41%	1.542	0.305
20 s temporal T	Transformer	58.26%	1.388	0.327
60 s temporal T	Transformer	58.77%	1.426	0.322
20 s patch T	Transformer	57.68%	1.447	0.331
60 s patch T	Transformer	59.20%	1.400	0.334
20 s hybrid T	Hybrid	57.63%	1.420	0.315
60 s hybrid T	Hybrid	58.91%	1.435	0.331

Preliminary fold-0 screen: one model per candidate; v4 combines 15.

Pure recipe	Top-1	Gain pp	Guards
Pure pair	62.95%	+0.87	yes
60 s patch T	62.23%	+0.15	no
60 s temporal T	62.20%	+0.11	no

Primary selection: quality guards passed on both partitions.

Each partition versus v4: NLL <= v4 +0.02; mean F1/33 >= v4 -0.01; each horizon accuracy >= v4 -0.003 (-0.30 pp). Secondary: mean accuracy gain >0 in both partitions.

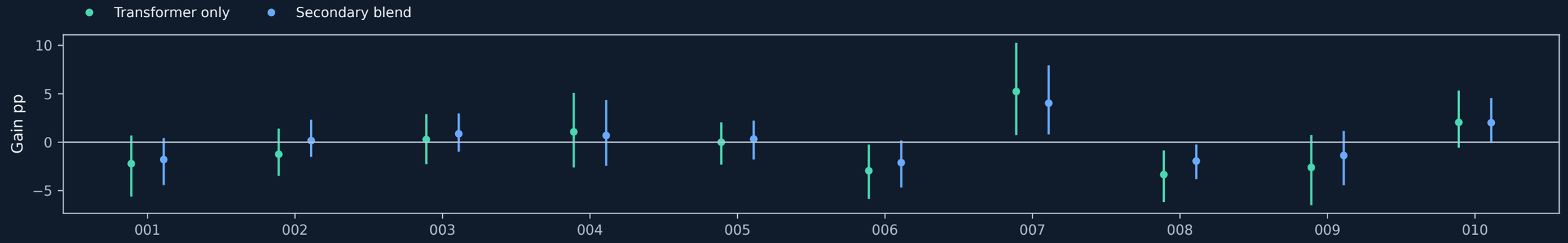
The final primary is a multi-seed ensemble: three seeds per shortlisted architecture, confirmed on both internal partitions against the cached 15-member v4 reference. These are internal selection scores.

Normalization uses fit groups; epochs and temperatures use stop groups; recipe ranking uses selection groups. The partitions overlap and have informed earlier studies. Final epochs, temperatures and reported outputs are frozen before validation.

+5 s: primary gains for 4 of ten pilots

Pilot point estimates and paired scenario uncertainty; the model is still evaluated on known trained policies

Pilot	Origins	v4 top-1	Primary	Gain pp	Secondary	Gain pp
001	2,845	55.85%	53.64%	-2.21	54.06%	-1.79
002	3,057	76.94%	75.70%	-1.24	77.10%	+0.16
003	2,878	65.46%	65.74%	+0.28	66.33%	+0.87
004	2,897	69.38%	70.45%	+1.07	70.07%	+0.69
005	3,049	74.52%	74.52%	+0.00	74.84%	+0.33
006	2,987	65.38%	62.44%	-2.95	63.27%	-2.11
007	3,001	54.88%	60.11%	+5.23	58.91%	+4.03
008	2,871	45.35%	42.01%	-3.34	43.40%	-1.95
009	2,990	71.77%	69.16%	-2.61	70.40%	-1.37
010	2,883	67.71%	69.75%	+2.05	69.72%	+2.01



Intervals are pointwise and conditional, without correction for checking ten pilots and three horizons. All 30 comparisons per model, including regressions, are in pilot_comparison.csv.

Per-skill F1: all 33 catalogue labels remain in scope

+5 seconds; support counts correlated eligible origins, not independent flights

Skill	n	v4 F1	Primary	Secondary
abort_support	113	0.515	0.506	0.525
accelerate_to_speed	0	—	—	—
beam_missile_left	6	0.000	0.000	0.000
beam_missile_right	20	0.513	0.513	0.526
beam_target_left	1,693	0.725	0.721	0.725
beam_target_right	1,445	0.737	0.724	0.733
climb_to_altitude	1	0.000	0.000	0.000
commit	0	—	—	—
crank_target_left	1,150	0.653	0.646	0.651
crank_target_right	947	0.788	0.776	0.778
decelerate_to_speed	126	0.307	0.284	0.304
defensive_reversal	1	0.000	0.000	0.000
descend_to_altitude	162	0.631	0.649	0.649
dive_defense	0	—	—	—
drag_missile	0	—	—	—
extend	2,676	0.602	0.584	0.597
lag_pursuit	4,872	0.632	0.638	0.640
last_ditch_break	0	—	—	—
launch	130	0.507	0.497	0.505
lead_pursuit	1,350	0.543	0.533	0.540
lock_target	0	—	—	—
maintain_heading	36	0.500	0.575	0.548
notch_left	124	0.430	0.420	0.442
notch_right	80	0.462	0.439	0.423
preemptive_crank	0	—	—	—
pursue_target	2,561	0.562	0.561	0.568
recommit	4,857	0.666	0.656	0.659
search	2	0.000	0.000	0.000
secondary_shot	1,414	0.656	0.651	0.658
short_range_attack	216	0.291	0.268	0.275
support_missile	19	0.429	0.160	0.160
turn_cold	5,437	0.701	0.706	0.707
turn_to_heading	20	0.585	0.500	0.531

Dash = no true examples at this horizon. Catalogue macro-F1 still averages all 33 labels; unsupported labels are not removed from the primary metric. Small-support changes need cautious interpretation.

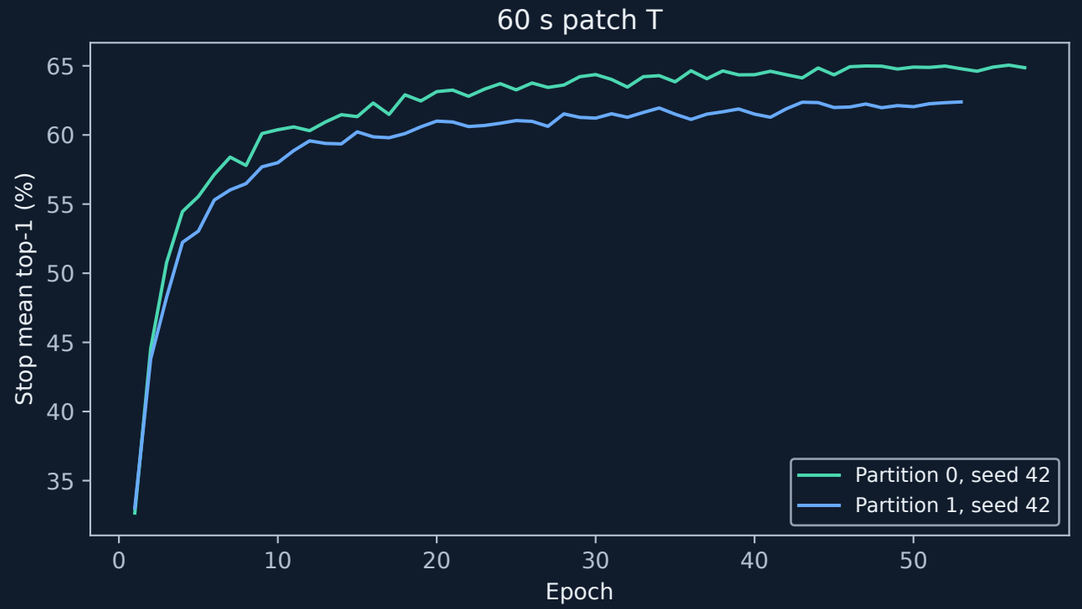
Full precision, recall, F1 and predicted counts for every pilot × horizon × skill, plus flight/scenario/pilot support, are exported. These labels measure selected catalogue behavior, not pilot competence.

Training cost, learning curves and reproducibility

All completed fits and rejected candidates are retained; reported curves are internal stop curves, not final-test curves

Phase	Runs	Logged time	Epochs
internal	23	112.73 min	1080
refit	6	22.57 min	210

29 new completed fits; 30 cached internal reference fits.
Summed logged time: 135.30 minutes. Loading/setup are excluded; shared-GPU execution, when used, adds contention.
This is not elapsed wall time or isolated throughput.



v4 incumbent: 15 models / 31.73M parameters.
Transformer only: 6 models / 18.91M parameters.
Secondary blend: 21 models / 50.64M parameters.

Shown: seed 42 separately in each partition, not a seed average. Final refits use fixed epochs and temperatures, without outer-validation callbacks. Training objectives, stop NLL and all seeds are exported.

Verified 927 preserved historical files, exact frozen source/protocol bytes, unchanged validation origins/masks and saved recipe probabilities.
Existing PDFs remain separate. No reserved test evaluation.